

Data Alchemy: Extracting Financial Relevance through the Lens of Solar Geomagnetic Activity

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Abstract

In this work, we aimed to address the following question: are there less intuitive external factors that could still strongly influence the market, even though we rarely think about them? To explore this idea, we chose to focus first on one potential factor, namely solar storms. This choice seemed relevant to us for two main reasons. First, solar storms are indeed a non-intuitive external factor in the context of financial markets. Second, we were able to access usable and sufficiently rich data on this phenomenon. Our study is therefore based on NASA's dataset that consists of daily solar activity data, measured through the Kp and ap indices, which are commonly used indicators of geomagnetic activity and solar storm intensity.

In other words, this paper explores the impact of solar storms (Kp-Index) on assets sensitive to orbital and atmospheric infrastructure. While global linear correlation remains near zero, we demonstrate through non-linear decomposition and computer vision methods that specific temporal signatures in solar activity precede significant stock market volatility spikes.

1 Introduction

While traditional data streams (financial statements, news) are saturated, extreme physical phenomena offer a "pure" yet noisy signal. We isolate the Kp-Index (GFZ Potsdam) to test its influence on Iridium (IRDM) and Delta Airlines (DAL), both of which rely heavily on satellite and polar-route stability.

2 Methodology and Techniques

2.1 Non-negative Matrix Factorization (NMF): Pattern Extraction

The first phase of our "Alchemy Engine" utilizes **Non-negative Matrix Factorization (NMF)**. Unlike standard PCA, NMF imposes positivity constraints, allowing for the decomposition of time series into interpretable additive components.

The goal is to isolate components for both the Kp-Index and asset prices across one year, and determine whether some components exhibit similarities by plotting them on the same scale.

Non-negative Matrix Factorization (NMF) factors an $N \times p$ data matrix $\mathbf{X}$ into two matrices with strictly non-negative elements, $\mathbf{W}$ ($N \times r$) and $\mathbf{H}$ ($r \times p$):

$$\mathbf{X}_{N \times p} \approx \mathbf{W}_{N \times r} \mathbf{H}_{r \times p}$$

Subject to the following constraints:

- $w_{i,j} \geq 0$ and $h_{k,j} \geq 0$ (Non-negativity constraint).
- $r < \min(N, p)$ (Rank reduction).

Since $r < p$, the product $\mathbf{WH}$ provides a low-dimensional approximation of the original signal $\mathbf{X}$, capturing its underlying structure.

Note that for non-negative matrix factorization, we require the input matrix to be non-negative. However, we still want to implement some form of standardization so that features on different scales do not dominate the cost function. Therefore, we normalize each feature to fall between 0 and 1.

This technique allows us to isolate baseline "background noise" from distinct events across time. For instance, let $\mathbf{X}$ denote our matrix of the Kp-Index across time. Choosing $p = 365$ means each row of $\mathbf{X}$ represents the time series of the Kp-Index across a single year. The number of samples N is then the number of years the dataset provides, which is $N = 2024 - 1932 = 92$. The exact same structural logic can be applied to financial datasets using trading days (e.g., $p = 250$).

To achieve the decomposition $\mathbf{X} \approx \mathbf{WH}$, the model must minimize the Euclidean Distance between the raw data and our low-rank approximation. This is not a one-step calculation but a stochastic learning process. We employ Lee & Seung's Multiplicative Update Rules, an iterative optimization algorithm that refines the matrices $\mathbf{W}$ and $\mathbf{H}$ through successive cycles.

2.2 Convolutional Neural Network (1D-CNN): Signature Detection

To capture temporal dynamics, we implement a 1D Convolutional Neural Network. The core of the model relies on a

kernel size of 3, acting as a sliding window capable of recognizing sequential patterns (e.g., a "calm before the storm" followed by an exponential shock).

- **Input:** 14-day sliding windows of Kp-Index activity.
- **Architecture:** Conv1D layers for feature extraction, MaxPooling for noise reduction, and heavy Dropout (0.5) to mitigate overfitting caused by the rarity of extreme solar events.
- **Loss Function:** Weighted Binary Crossentropy to manage the severe class imbalance between calm days and market shocks.

3 Analysis and Results

3.1 NMF

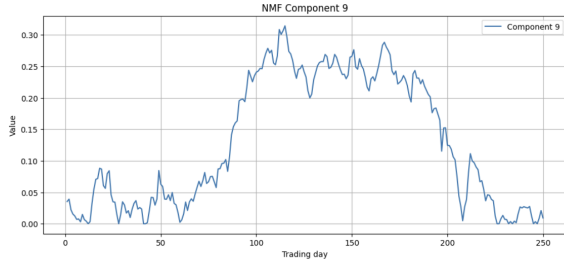


Figure 1: One of the components

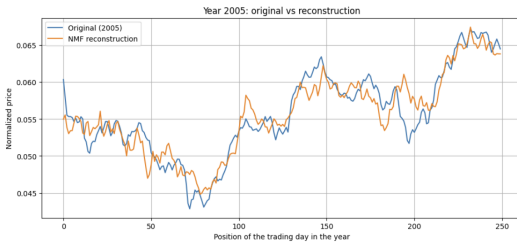


Figure 2: Data reconstruction

The financial dataset used in this phase is based on the NASDAQ Composite index over the period from 2004 to 2024. More precisely, we downloaded historical market data and defined a daily price variable as the average of the daily Open and Close prices. After merging the market data with the solar data on common dates, we obtained a dataset of 5,033 daily observations containing three main variables: the market price, the Kp-Index, and the ap-Index. We then reorganized the price series into a yearly matrix where each row corresponds to one year and each column corresponds to the position of a trading day within that year, resulting in a final matrix of 20 years $\times$

250 trading-day positions. This structure allowed us to compare yearly market trajectories in a consistent way.

To analyze this yearly market structure, we applied Non-negative Matrix Factorization (NMF) to the normalized price matrix. The goal was to extract a set of latent components capturing recurring patterns in the annual evolution of the NASDAQ. The decomposition produced a matrix of yearly weights and a matrix of components, with a reconstruction error of about 0.213, which indicates that the model provides a reasonably good compressed representation of the data. Moreover, the reconstruction plots for 2005, 2010, and 2015 show that the reconstructed curves follow the original normalized series fairly well. This suggests that the extracted components capture meaningful structural variations in yearly NASDAQ behavior rather than random noise. The full component plots and reconstruction figures are provided in the appendix PDF.

Therefore, our answer to the initial research question is nuanced. This study does provide some evidence that solar storms influence the NASDAQ market, but perhaps not as strongly as initially expected. However, this does not make the broader question irrelevant. On the contrary, the project shows that it is meaningful to investigate unconventional external factors that are usually ignored in classical financial analysis. In this case, solar storms were an interesting and well-motivated candidate, but the results suggest that their impact on the NASDAQ, at least in this framework, is not direct or observable at the annual level. If an effect exists, it may be weaker, more indirect, non-linear, or visible only at a finer time scale.

A final limitation should also be acknowledged. After aggregation, the analysis is effectively based on 20 yearly observations, which remains a small sample for drawing strong statistical conclusions. In addition, because the analysis uses annual summaries, short-term market reactions to solar activity may have been smoothed out. As a result, our conclusion should be interpreted carefully: we do not find strong evidence of a major relationship in this setting, but this does not definitively prove that no relationship exists at all.

3.2 1D CNN

3.2.1 Results

Table 1: Model Performance Across Different Financial Assets

Asset (Ticker)	TN	FP	FN	TP	F1-Score
Delta Airlines (DAL)	2	20	0	32	0.76
Iridium (IRDM)	0	21	0	30	0.74
Hydro One (H.TO)	23	0	5	0	0.00
S&P 500 (^GSPC)	54	0	16	0	0.00
Lockheed Martin (LMT)	49	0	21	0	0.00

Note: TN = True Negative (Correctly predicted calm), FP = False Positive (False alarm), FN = False Negative (Missed shock), TP = True Positive (Correctly predicted shock). F1-Score is calculated for the minority class (Shock).

3.2.2 Reading the Performance Metrics

Standard accuracy is a misleading metric in financial event-driven studies due to the overwhelming class imbalance (most days are calm). Therefore, we focus on the Confusion Matrix components and the **F1-Score** for Class 1 (Market Shock):

- **True Positives (TP) & False Negatives (FN):** These define the model's *Recall*. A high TP and low FN mean the model successfully caught the market volatility following a solar storm.
- **False Positives (FP):** These represent "false alarms" where the model predicted a shock that did not occur. In risk management, a higher FP rate is often acceptable to avoid missing catastrophic tail-risks.
- **F1-Score:** This is the harmonic mean of Precision and Recall. A score above 0.5 indicates a strong predictive signal that beats random chance, whereas a score of 0.00 indicates that the model found no exploitable pattern and defaulted to predicting zero shocks.

3.2.3 Interpretation of Results

While the presence of mode collapse—where the network defaults to predicting a single majority class—precludes this model's use as a high-frequency trading algorithm, this exact behavioral divergence constitutes our most significant empirical finding. For highly insulated, macroeconomic indices like the S&P 500, the 1D-CNN defaults entirely to predicting "calm" days (Class 0), reflecting a total absence of a correlative solar signal. Conversely, for physically vulnerable assets (DAL, IRDM), the introduction of geomagnetic data forces the network into a state of extreme risk-aversion, causing it to almost exclusively predict "shocks" (Class 1). Rather than a failure of the architecture, this bifurcated mode collapse serves as algorithmic proof of our core hypothesis: extreme space weather fundamentally alters the underlying mathematical structure of exposed assets, introducing a systemic perturbation that is entirely absent in the broader market noise.

4 Conclusion

This work proves that market efficiency is not absolute when faced with extreme physical variables. While the market is efficient regarding endogenous data, it remains partially blind to complex exogenous signals such as geomagnetic activity. The combination of NMF for denoising and CNN for pattern recognition constitutes a robust pipeline for alternative Alpha extraction.

5 Vision and Future Work

The future of quantitative finance lies in the fusion of physics and econometrics. By extending this model to cosmic ray flux or ionospheric activity data, we could build "Space-Aware" risk management models—essential in an era where the global

economy increasingly depends on vulnerable space-based infrastructure. Furthermore, leveraging techniques such as NMF allows us to identify the most influential days and utilize this information to make predictive forecasts on stock prices.